

# Ligjerata 8c

Ligji i Arkimedit

$$F_A = F_2 - F_1$$

$$F_1 = p_1 S = \rho_f g h_1 \cdot S$$

$$F_2 = p_2 S = \rho_f g h_2 \cdot S$$

$$F_A = \rho_f g h_2 S - \rho_f \cdot g h_1 S$$

$$F_A = \rho_f g (h_2 - h_1) \cdot S$$

$$F_A = \rho_f g \underbrace{h \cdot S}_V \quad ; \quad h = h_2 - h_1$$

$$F_A = \rho_f \cdot g V$$

$$V \cdot \rho_f = m_f$$

$$F_A = m_f g$$

$$F_f = \rho_f \cdot g V$$

$$Q = m g = \rho \cdot g V$$

$$F = F_A - Q$$

$$F = \rho_f \cdot g V - \rho g V$$

$$F = g V (\rho_f - \rho)$$

$$\textcircled{1} \quad \rho_f > \rho_{\text{trupit}}$$

$$F_A > Q$$

$$F = g V (\rho_f - \rho) > 0$$

$$\rho_f g V = \rho g V_0$$

$$\rho_f V = \rho V_0$$

$$m_f = m_0$$

$$\textcircled{2} \quad \rho_f = \rho_{\text{trupit}}$$

$$F = g V (\rho_f - \rho) = 0$$

$$F_A = Q$$

$$\textcircled{3} \quad \rho_f < \rho_{\text{trupit}}$$

$$F = g V (\rho_f - \rho) < 0$$

$$F_A < Q$$

$$Q_a = m_a \cdot g$$

$$Q_u = m_u g$$

$$\gamma = \rho \cdot g$$

$$Q_a - Q_u = \gamma V$$

$$m_a g - m_u g = \rho V \cdot g$$

$$\rho_u = \frac{m_a - m_u}{V}$$

